Session 5 · Deck 5.A3

Which Solution, Higher Order, and Distributions

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Thursday, October 22, 2026

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Where We Are

you are here: 5.A3 ▼

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Block A — perturbation: 5.A1 the idea · 5.A2 the growth model to first order · **5.A3** which solution, higher order, distributions.

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Roadmap: Three Parts

Three things the first-order solution leaves open: which of its solutions is the economy's; what it cannot see; and whether the expansion point has to be a point.

Part	Topic	The question it answers
1	Blanchard–Kahn	The linear system has several solutions. Which is the economy's?
2	Second order: where risk lives	First order is certainty equivalent. What does one more order buy?
3	From a point to a distribution	Can the recipe handle a state that is a whole distribution?

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Idea

Part 1 is step 6 of 5.A1's table and the only genuinely new tool in Block A. Parts 2 and 3 are the same recipe — differentiate the identity, match coefficients — pushed one order further and one state further.

I. Two Kinds of Variables, One Standard Form

Blanchard–Kahn: a linear rational-expectations system has a unique stable (saddle-path) solution if and only if the number of roots with modulus above 1 equals the number of jump variables. First, sort the variables.

	Predetermined (state) x_t	**Jump** (forward-looking) y_t
meaning	set by history at the start of t	adjusts at t to news; looks forward
RBC	capital $\hat{k}_t$, TFP z_t	consumption $\hat{c}_t$ (or labour, investment)
count	$n_1 = 2$	$n_2 = 1$

After linearizing (Step 1), with M of size $(n_1 + n_2) \times (n_1 + n_2)$:

$$\begin{pmatrix} x_{t+1} \\ \mathbb{E}_t\, y_{t+1} \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \begin{pmatrix} 0 \\ \sigma\varepsilon_{t+1} \\ 0 \end{pmatrix}.$$

Why no $\mathbb{E}_t$ on the x rows

A predetermined variable has no expectation error: $\hat{k}_{t+1}$ is chosen at t. The shock enters directly: z_{t+1} is ρz_t plus the new $\sigma\varepsilon_{t+1}$.

Idea

History gives x_0. Nothing gives y_0: each $\hat{c}_0$ starts a different path. Blanchard–Kahn says when exactly one of them stays bounded.

Blanchard & Kahn (1980), *Econometrica* 48(5). Hats: percent (log) deviations from the steady state.

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II. The Blanchard–Kahn Condition

Count the roots ω of M by modulus: m_1 stable ($|\omega| < 1$), m_2 unstable ($|\omega| > 1$).

Roots vs. jumps	The solution	Economic meaning
$m_2 = n_2$ (so $m_1 = n_1$)	**unique**, stable: the saddle-path equilibrium	BK holds: for any x_t, exactly one y_t puts the economy on the saddle path back to the steady state
$m_2 > n_2$	**none**: explosive	more unstable directions than jump variables can control; almost any deviation diverges, violating transversality
$m_2 < n_2$	**many**: indeterminacy	too few anchors for the jump variables; infinitely many paths converge: sunspot equilibria

Intuition

Each root above 1 carries a term that grows like $\omega^t \to \infty$. Transversality forbids it. The only freedom agents have is the jump variables chosen at t; they use it to set each explosive coefficient exactly to zero. n_2 jump variables cancel exactly n_2 explosive roots.

Example: the growth model

$|\omega| = 0.950,\ 0.962 \mid 1.050$: $m_2 = 1 = n_2$. **Unique.**

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Ledger `blanchard_kahn`. 2.B2's OLG indeterminacy (Feng & Hoelle 2017) is the third row.

III. Step 1: Steady State, Then Log-Linearize

Set every variable at its steady state ($z = 0$) and solve; then take a first-order expansion of the three dynamic equations around it, and stack them.

Steady state. $\beta R^* = 1$ gives k^*; then $y^* = k^{*\alpha}$, $c^* = y^* - \delta k^*$.

The three linearized equations (CRRA γ; $\gamma = 1$ here):

1. **Euler:** $\mathbb{E}_t \hat{c}_{t+1} - \hat{c}_t = \frac{1}{\gamma}\,\mathbb{E}_t \hat{R}_{t+1}$, with $\hat{R}_{t+1} = \kappa\big(z_{t+1} + (\alpha - 1)\hat{k}_{t+1}\big)$, a combination of $\hat{k}_{t+1}$ and z_{t+1}.
2. **Capital accumulation:** $\hat{k}_{t+1} = (1-\delta)\hat{k}_t + \dfrac{y^*}{k^*}\hat{y}_t - \dfrac{c^*}{k^*}\hat{c}_t$, with $\hat{y}_t = z_t + \alpha \hat{k}_t$.
3. **Shock:** $z_{t+1} = \rho z_t + \sigma \varepsilon_{t+1}$.

Stack them in matrix form: the standard form of Part I, with $x = (\hat{k}, z)$, $y = \hat{c}$.

Example: quarterly numbers

$k^* = 28.348$, $y^* = 3.015$, $c^* = 2.307$, so $y^*/k^* = 0.106$, $c^*/k^* = 0.081$, and $\kappa = \beta\alpha k^{*\alpha-1} = 0.0348$.

In detail

The next four frames derive each line: what linearizing means, the capital equation term by term, the Euler equation step by step, and the stacked matrices.

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Ledger `qz_detail.steady_state; pert_first_order.kappa`.

1· Step 1 in Detail (a): Linearizing Is a First-Order Taylor Expansion

Yes: linearizing is a first-order Taylor expansion of each equilibrium condition, with $\hat{c}_t$ kept as a variable of its own.

A condition $H(q) = 0$, where q lists today's and tomorrow's variables, holds at the steady state q^*. Three steps:

1. **Taylor to first order** around q^*: $H(q) \approx H(q^*) + \sum_q \left(\frac{\partial H}{\partial q}\right)^* (q - q^*)$.
2. **The constant is zero**, because the condition holds at the steady state: $H(q^*) = 0$.
3. **Measure deviations in percent**: $\hat{q} = \ln q - \ln q^*$, and $q - q^* \approx q^* \hat{q}$ (Taylor again, of $\ln q$). Together:

$$\sum_{\text{variables } q} q^* \left(\frac{\partial H}{\partial q}\right)^* \hat{q} \;=\; 0.$$

Example: one term

$e^z k^\alpha$: $\partial/\partial z = y^*$, $k^* \partial/\partial k = \alpha y^*$, so $e^z k^\alpha \approx y^*(1 + z + \alpha \hat{k})$.

Idea: two roads, four numbers

Road 1 (5.A2): substitute the linear policy and match coefficients — a quadratic, several solutions. **Road 2 (here):** linearize first, then ask which *paths* stay bounded. The bounded path *is* the policy, and it picks Road 1's solution.

Units

Hats are now **log deviations** (5.A2 used levels). Slopes rescale; eigenvalues do not.

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Ledger `pert_first_order`: **the two roads give the same four coefficients to** 4×10^{-11}.

1 · Step 1 in Detail (b): Capital Accumulation, Term by Term

The model is $c_t + k_{t+1} = e^{z_t} k_t^{\alpha} + (1-\delta)k_t$ with CRRA utility at $\gamma = 1$ ($\ln c$). Expand each term to first order; the steady-state parts cancel.

Term	Level part	Deviation part
c_t	c^*	$c^* \hat{c}_t$
k_{t+1}	k^*	$k^* \hat{k}_{t+1}$
$e^{z_t} k_t^{\alpha}$	y^*	$y^* z_t + \alpha y^* \hat{k}_t$
$(1-\delta)k_t$	$(1-\delta)k^*$	$(1-\delta)k^* \hat{k}_t$

Level column: the steady state itself; it cancels. **Deviation column** is what is left:

$$c^* \hat{c}_t + k^* \hat{k}_{t+1} \;=\; y^* z_t + \big(\alpha y^* + (1-\delta)k^*\big)\hat{k}_t.$$

Divide by k^*, with $\hat{y}_t = z_t + \alpha \hat{k}_t$: Step 1's line 2,

$$\hat{k}_{t+1} = (1-\delta)\hat{k}_t + \tfrac{y^*}{k^*}\hat{y}_t - \tfrac{c^*}{k^*}\hat{c}_t.$$

Idea

Each term becomes (its steady-state level) $\times$ (its percent deviation). The levels cancel because the steady state satisfies the constraint. Only the deviation part is the linear equation.

Example: quarterly numbers

$k^* = 28.348$, $y^* = 3.015$, $c^* = 2.307$:

$$2.307\hat{c}_t + 28.348\hat{k}_{t+1} = 3.015 z_t + 28.635\hat{k}_t$$

Ledger `qz_detail`, **keys** `steady_state`, `A`, `B`.

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1· Step 1 in Detail (c): The Euler Equation, and the Shock

$\frac{1}{c_t} = \beta\,\mathbb{E}_t\frac{R_{t+1}}{c_{t+1}}$, **with** $R_{t+1} = \alpha e^{z_{t+1}} k_{t+1}^{\alpha-1} + 1 - \delta$ **and** $\beta R^* = 1$ **at the steady state. Same first-order rule, four steps.**

1. **Left side.** $1/c_t = e^{-\ln c_t} \approx \frac{1}{c^*}(1 - \hat{c}_t)$.
2. **Right side.** $\dfrac{R_{t+1}}{c_{t+1}} \approx \dfrac{R^*}{c^*}\big(1 + \hat{R}_{t+1} - \hat{c}_{t+1}\big)$.
3. **Equate**, use $\beta R^* = 1$, cancel $1/c^*$: $1 - \hat{c}_t = \mathbb{E}_t\big[1 + \hat{R}_{t+1} - \hat{c}_{t+1}\big]$, so
$$\mathbb{E}_t\hat{c}_{t+1} = \hat{c}_t + \mathbb{E}_t\hat{R}_{t+1}.$$
4. **The return.** Only the marginal product moves ($1 - \delta$ is a constant): $R_{t+1} - R^* \approx \alpha k^{*\alpha-1}\big(z_{t+1} + (\alpha-1)\hat{k}_{t+1}\big)$. Divide by R^*: $\hat{R}_{t+1} = \kappa\big(z_{t+1} + (\alpha-1)\hat{k}_{t+1}\big)$.

Euler row: $\mathbb{E}_t\hat{c}_{t+1} - \kappa\,\mathbb{E}_t z_{t+1} - \kappa(\alpha-1)\hat{k}_{t+1} = \hat{c}_t$.

Shock row: $\mathbb{E}_t z_{t+1} = \rho z_t$ — already linear, nothing to do.

Where κ comes from

$\kappa = \alpha k^{*\alpha-1}/R^* = \beta\alpha k^{*\alpha-1}$: the marginal product's **share of the gross return**. Quarterly, $\kappa = 0.0348$; at $\delta = 1$, $\kappa = 1$.

Check at $\delta = 1$

The exact policy is $\hat{c}_t = \hat{k}_{t+1} = z_t + \alpha\hat{k}_t$ (5.A2), so $\mathbb{E}_t\hat{c}_{t+1} = \rho z_t + \alpha\hat{k}_{t+1}$. The Euler row gives $\hat{k}_{t+1} + \rho z_t + (\alpha - 1)\hat{k}_{t+1}$: the same.

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Ledger pert_first_order, **key** kappa.

1· Step 1 in Detail (d): Stack Them — A, B, and M

Put the three linearized equations into one matrix system $A\,\mathbb{E}_t s_{t+1} = B\,s_t$ with $s_t = (\hat{k}_t,\, z_t,\, \hat{c}_t)^\top$, then isolate tomorrow on the left.

1. Capital:
2. Shock:
3. Euler:

$$\underbrace{\begin{pmatrix} k^* & 0 & 0 \\ 0 & 1 & 0 \\ -\kappa(\alpha-1) & -\kappa & 1 \end{pmatrix}}_{A \text{ (weights on tomorrow)}} \mathbb{E}_t \begin{pmatrix} \hat{k}_{t+1} \\ z_{t+1} \\ \hat{c}_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} \alpha y^* + (1-\delta)k^* & y^* & -c^* \\ 0 & \rho & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{B \text{ (weights on today)}} \begin{pmatrix} \hat{k}_t \\ z_t \\ \hat{c}_t \end{pmatrix}$$

From $A\,\mathbb{E}_t s_{t+1} = B s_t$ to $\mathbb{E}_t s_{t+1} = M s_t$:

- **Row 1 ($\hat{k}_{t+1}$):** divide by k^* to get $\hat{k}_{t+1} = 1.010\hat{k}_t + 0.106 z_t - 0.081\hat{c}_t$.
- **Row 2 (z_{t+1}):** already isolated, $\mathbb{E}_t z_{t+1} = 0.950\, z_t$.
- **Row 3 ($\mathbb{E}_t \hat{c}_{t+1}$):** plug Rows 1 and 2 into the Euler row $\mathbb{E}_t \hat{c}_{t+1} = \hat{c}_t + \kappa(\alpha-1)\hat{k}_{t+1} + \kappa\,\mathbb{E}_t z_{t+1}$.
- In matrix form, that substitution is $M = A^{-1}B$.

Standard form $M = A^{-1}B$, split by $(x_t \mid y_t)$

$$M = \left(\begin{array}{cc|c} 1.010 & 0.106 & -0.081 \\ 0 & 0.950 & 0 \\ \hline -0.024 & 0.031 & 1.002 \end{array}\right)$$

Top: M_{11}, M_{12} (state $x_t = (\hat{k}_t, z_t)$ rows);
Bottom: M_{21}, M_{22} (jump $y_t = \hat{c}_t$ row).

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Ledger bk_blocks.M; qz_detail.A, B.

1· Step 2: Decompose M Into Stable and Unstable Roots

Why can't we solve $\mathbb{E}_t s_{t+1} = M s_t$ by inspection? Because $\hat{k}_t$, z_t, and $\hat{c}_t$ are tangled inside M. We diagonalize M to find its pure growth rates (eigenvalues ω_i).

1. **Find the roots and eigenvectors of M.** Stack the eigenvectors in V and eigenvalues in $\Omega = \mathrm{diag}(\omega_1, \omega_2, \omega_3)$: $MV = V\Omega$.
2. **Invert V to isolate Ω.** Set $L \equiv V^{-1}$:
$$M = V\Omega L \quad \Longleftrightarrow \quad LM = \Omega L.$$
Multiplying our system on the left by L will turn the tangled matrix M into the diagonal Ω (Step 3).
3. **Sort by stability and split L into blocks** (m_1 stable $|\omega| < 1$, m_2 unstable $|\omega| > 1$):
$$\Omega = \begin{pmatrix} \Omega_1 & 0 \\ 0 & \Omega_2 \end{pmatrix}, \qquad L = \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{pmatrix} \begin{matrix} \\ \leftarrow m_1 \text{ stable} \\ \leftarrow m_2 \text{ unstable} \end{matrix}$$

Example: the growth model

Roots: $\Omega = \mathrm{diag}(\underbrace{0.950,\ 0.962}_{\Omega_1 \text{ (stable)}} \mid \underbrace{1.050}_{\Omega_2 \text{ (unstable)}})$.

With columns $(\hat{k}_t, z_t \mid \hat{c}_t)$:

$$L = \left(\begin{array}{cc|c} 0 & 1 & 0 \\ 0.453 & 6.342 & 0.926 \\ \hline 0.547 & 0.299 & -0.926 \end{array}\right).$$

Bottom (unstable) row for $\omega_3 = 1.050$: $L_{21} = (0.547,\ 0.299)$ on x_t and $L_{22} = -0.926$ on y_t.

In practice Dynare uses the generalized Schur (QZ) decomposition on (A, B) directly. Ledger `bk_blocks: Omega_diag, L`.

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1· Step 3: Change Coordinates — Untangle Into Independent Modes

Multiply $\mathbb{E}_t s_{t+1} = M s_t$ by L and define $\tilde{s}_t \equiv L s_t$: because $LM = \Omega L$, the tangled 3×3 system splits into separate 1-variable equations.

1. **Pre-multiply the system by L:**
$$L\,\mathbb{E}_t s_{t+1} = LM s_t = \Omega\,(L s_t) \implies \mathbb{E}_t \tilde{s}_{t+1} = \Omega\,\tilde{s}_t.$$

2. **Split $\tilde{s}_t \equiv L s_t$ into stable ($\tilde{x}_t$) and explosive ($\tilde{y}_t$):**
$$\begin{pmatrix} \tilde{x}_t \\ \tilde{y}_t \end{pmatrix} = \begin{pmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} L_{11}x_t + L_{12}y_t \\ L_{21}x_t + L_{22}y_t \end{pmatrix}.$$

3. **Each block now evolves on its own:**
$$\mathbb{E}_t \tilde{x}_{t+1} = \Omega_1\,\tilde{x}_t \quad (|\Omega_1| < 1\text{: } \textbf{shrinks} \text{ to } 0)$$
$$\mathbb{E}_t \tilde{y}_{t+1} = \Omega_2\,\tilde{y}_t \quad (|\Omega_2| > 1\text{: } \textbf{explodes} \text{ if } \tilde{y}_t \neq 0)$$

Example: the 3 untangled modes

Multiplying the rows of L by $(\hat{k}_t, z_t, \hat{c}_t)^\top$:

$$\tilde{x}_{1,t} = z_t \qquad (\times 0.950)$$
$$\tilde{x}_{2,t} = 0.453\hat{k}_t + 6.342 z_t + 0.926\hat{c}_t \quad (\times 0.962)$$
$$\tilde{y}_t = 0.547\hat{k}_t + 0.299 z_t - 0.926\hat{c}_t \quad (\times \mathbf{1.050})$$

Each mode evolves alone: $\mathbb{E}_t \tilde{s}_{i,t+1} = \omega_i \tilde{s}_{i,t}$.

Why this makes the problem easy

$\tilde{x}_{1,t}$ and $\tilde{x}_{2,t}$ decay on their own, while **all** explosive behavior is trapped inside a single number, $\tilde{y}_t$.

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Ledger bk_blocks: L11, L12, L21, L22.

1· Step 4: Kill the Explosive Mode — Why y_t Does It

The procedure, at any date t. Only the jump variable is still free, so it is the one that sets $\tilde{y}_t = 0$.

1. **Take the state as given.** $\hat{k}_t$ was chosen at $t-1$; nature has drawn z_t. So x_t is fixed; only $y_t = \hat{c}_t$ is open.
2. **Write the explosive mode,** $\tilde{y}_t = L_{21}x_t + L_{22}y_t$. The model fixes its future, $\mathbb{E}_t\tilde{y}_{t+T} = \Omega_2^T\tilde{y}_t$; nothing chosen later undoes it.
3. **Require it to be exactly zero.** Otherwise it grows like Ω_2^T: capital is eaten away or piles up without bound, violating transversality (phase diagram, below).
4. **Solve for y_t.** x_t cannot move, so y_t must:
$$L_{21}x_t + L_{22}y_t = 0 \implies y_t = \underbrace{-L_{22}^{-1}L_{21}}_{C}\, x_t.$$
5. **Repeat next date.** The shock moves x_{t+1}; y_{t+1} jumps again so that $\tilde{y}_{t+1} = 0$. Same C every date.

Why the jump variable

History fixes x_t. The Euler equation ties how consumption *changes* from t to $t+1$, not its level today. That level is one free number, used to set one explosive mode to zero: $\hat{c}$ moves at once on news; $\hat{k}$ cannot.

Where $m_2 = n_2$ lives

$\tilde{y}_t = 0$ is m_2 equations in n_2 unknowns. L_{22} ($m_2 \times n_2$) is invertible only if $m_2 = n_2$ and $\det L_{22} \neq 0$.

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1 · Step 4, Worked: The Quarter the Shock Hits

TFP rises by one standard deviation; capital has not moved. Follow the procedure line by line.

1. **The state.** $\hat{k}_0 = 0$ (given), $z_0 = \sigma = 0.007$.
2. **The explosive mode, for any $\hat{c}_0$** (bottom row of L):
   $\tilde{y}_0 = 0.547{\cdot}0 + 0.299{\cdot}0.007 - 0.926\,\hat{c}_0 = 0.00209 - 0.926\,\hat{c}_0$.
3. **Kill it.** $\hat{c}_0 = 0.00209/0.926 = \mathbf{0.226}\%$, which is $c_z\sigma = 0.323 \times 0.7\%$.
4. **Any other $\hat{c}_0$ explodes.** At $\hat{c}_0 = 0.30\%$, $\tilde{y}_0 = -0.00069$, and it grows $\times 1.050$ a quarter ($\times 7.0$ in 40 quarters). Expected paths, in %:

quarter	0	20	40	80
$\hat{k}$, $\hat{c}_0 = 0.226\%$	0	0.48	0.39	0.13
$\hat{k}$, $\hat{c}_0 = 0.30\%$	0	0.33	-0.07	-3.24

5. Next quarter: the same step again

$x_1 = Kx_0 = (0.080 \times 0.007,\ 0.950 \times 0.007) = (0.00056,\ 0.00665)$. Then $\hat{c}_1 = Cx_1 = 0.590 \times 0.00056 + 0.323 \times 0.00665 = 0.248\%$, again the only value with $\tilde{y}_1 = 0$.

Key intuition

Too much consumption today eats the capital stock: $\hat{k}$ turns negative and keeps falling. Repeating steps 1–4 every quarter traces the impulse response (Lab 5, Parts 2–3, run these numbers).

Ledger `bk_jump_example: c0_star, ytilde0_wrong, paths, x1, c1;` C **agrees with** `bk_blocks.`

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1· Step 5: Plug the Policy Back In — the State Law of Motion

Plug the jump rule $y_t = Cx_t$ back into the state rows of M to get the state law of motion $x_{t+1} = Kx_t +$ shock.

1. **Eliminate y_t from the state block.** Put $y_t = Cx_t$ into $x_{t+1} = M_{11}x_t + M_{12}y_t + (0, \sigma\varepsilon_{t+1})^\top$:

$$x_{t+1} = \underbrace{(M_{11} + M_{12}C)}_{K} x_t + \begin{pmatrix} 0 \\ \sigma\varepsilon_{t+1} \end{pmatrix}.$$

2. **In the growth model.** Plug $\hat{c}_t = 0.590\hat{k}_t + 0.323z_t$ into $\hat{k}_{t+1} = 1.010\hat{k}_t + 0.106z_t - 0.081\hat{c}_t$ to get $\hat{k}_{t+1} = (1.010 - 0.048)\hat{k}_t + (0.106 - 0.026)z_t$:

$$K = \begin{pmatrix} 0.962 & 0.080 \\ 0 & 0.950 \end{pmatrix} = \begin{pmatrix} k_k & k_z \\ 0 & \rho \end{pmatrix}.$$

Economic intuition: why $k_k = 0.962 < 1$

Without consumption feedback ($\hat{c}_t = 0$), extra capital compounds at $1/\beta = 1.010 > 1$. Consuming part of the extra output ($c_k = 0.590 > 0$) pulls the capital root down to $0.962 < 1$.

Summary: what the two sets of roots do

Unstable (Ω_2): set $\tilde{y}_t = 0$ to pin down the policy C.
Stable (Ω_1): become the eigenvalues of the transition K.

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Ledger bk_blocks: M11, M12_C, K, eig_K; eigen_route.start.

1· IV. The Same Answer as Undetermined Coefficients

In 5.A2 we solved the model without matrices by guessing a linear policy and matching coefficients. Why does that give a quadratic, and why the same answer?

1. **Guess linear rules** ($z_t = 0$): $\hat{c}_t = c_k \hat{k}_t$ and $\hat{k}_{t+1} = k_k \hat{k}_t$, which implies $\hat{c}_{t+1} = c_k \hat{k}_{t+1} = c_k k_k \hat{k}_t$.
2. **Plug the guess into the two linearized equations:**
   - *Capital* ($\hat{k}_{t+1} = 1.010\hat{k}_t - 0.081\hat{c}_t$): $k_k = 1.010 - 0.081\, c_k$.
   - *Euler* ($\hat{c}_{t+1} + 0.023\hat{k}_{t+1} = \hat{c}_t$, since $-\kappa(\alpha - 1) = 0.023$): $c_k k_k + 0.023\, k_k = c_k$.
3. **Why a quadratic.** Substituting $c_k = (1.010 - k_k)/0.081$ into $c_k k_k$ gives a quadratic in k_k. Its two roots are the **exact eigenvalues** from Step 2: $k_k \in \{\mathbf{0.962}, 1.050\}$.
4. **Pick the stable root.** Keeping $k_k = 0.962 < 1$ gives $c_k = 0.590$ (and matching z_t gives $c_z = 0.323$, $k_z = 0.080$). Discarding $1.050 > 1$ is identical to setting $\tilde{y}_t = 0$ in Step 4!

Both calibrations: two roots for k_k

	keep ($k_k < 1$)	discard (> 1)
$\delta = 1$ (5.A2)	$\alpha = 0.360$	2.924
quarterly	0.962	1.050

Why use Steps 2–5 in practice?

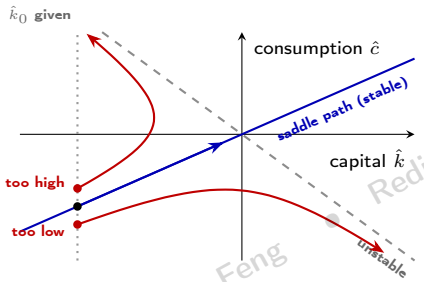
With one jump variable, undetermined coefficients gives a scalar quadratic. With many variables, it gives coupled matrix quadratics — solved in one decomposition by Steps 2–5.

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Ledger `blanchard_kahn` **(both calibrations)**; `qz_detail.kc_block`; `pert_first_order`: **the two roads agree to** 4×10^{-11}.

1· The Phase Diagram: Why Only the Saddle Path Converges

In the $(\hat{k}, \hat{c})$ plane the stable root draws one line back to the steady state; the unstable root carries every other start away.



Schematic, $z = 0$. $\hat{k}_0$ is given; only the black $\hat{c}_0$ lies on the saddle path.

- **Saddle path:** the stable direction ($\omega = 0.962$), $\hat{c} = c_{\hat{k}}\hat{k}$ with $c_{\hat{k}} = 0.590$. On it, the deviation shrinks by 0.962 a quarter.
- **Unstable direction** ($\omega = 1.050$): slope -0.490 in $(\hat{k}, \hat{c})$.
- **Off the path**, the start has weight on the unstable direction, and 1.050^t dominates. Too high: consumption eats the capital stock. Too low: capital piles up without bound, violating transversality.
- **Change the parameters** and both roots move, but their product stays $1/\beta > 1$, so they cannot both be stable; in the growth model one is always below 1, and the saddle path is unique.

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Ledger `eigen_route.V_columns_scaled`: **directions** $(1, 0, 0.590)$ **and** $(1, 0, -0.490)$ **in** $(\hat{k}, z, \hat{c})$; `qz_detail.kc_block`.

The first-order solution is certainty equivalent: the same in a risky economy and a riskless one. Whatever it says about uncertainty was assumed, not computed.

5.A2 showed $c_\chi = k_\chi = 0$: the linear terms in the shock size cancel. So at first order there is

- no **precautionary saving**,
- no **risk premium**, no welfare cost of uncertainty.

Second and higher orders restore them:

- second order: precaution through $c_{\chi\chi}$, risk premia, welfare costs;
- third order: effects of **skewness** and kurtosis of the shocks.

The cost is algebra — and the method is still **local** (Part 2's last frame).

Idea

A straight line cannot bend. Precaution is about how marginal utility *curves* (2.A1's prudence, $u''' > 0$), and curvature is a second-derivative object. So it can only appear at second order.

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Second Order: Twelve Equations, Twelve Unknowns, and Linear

Differentiate $F = 0$ twice. Six second derivatives, each a 2-vector, give twelve equations; the first-order coefficients are now known numbers, so the new unknowns enter linearly.

At $(k^*, 0; 0)$:

$F_{kk} = F_{kz} = F_{k\chi} = F_{zz} = F_{z\chi} = F_{\chi\chi} = 0.$

Unknowns: six second-order coefficients of c, six of k: **12**. **Equations:** six conditions $\times$ two components: **12**.

Linear: with the first-order coefficients known, the new unknowns appear only to the first power. The policy gains

$$\tfrac{1}{2}\, c_{\chi\chi}\, \chi^2 \quad \text{at } \chi = 1: \quad \text{a } \textbf{constant} \text{ shift},$$

the precautionary term: consumption below the deterministic level, capital above it, because of risk alone.

Idea

5.A1's chain-rule frame, one order up: each order is linear given the orders below. Third order is 20 more equations of the same kind. **The procedure is recursive** and can be continued as far as you like.

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Certainty equivalence is gone: the policy depends on σ, and the economy fluctuates around a *risk-adjusted* point.

Perturbation: Strengths and Limits

Fast, with free impulse responses and moments, and it scales to many states. Local, so it fails far from the steady state, at a kink, and for anything that lives in the tails.

What perturbation does well

- Very fast: a steady state, derivatives, one decomposition — exact to the order solved when the derivatives are analytic.
- Impulse responses and moments come almost free.
- Scales to many state variables — even a whole distribution (Part 3).

What it cannot do

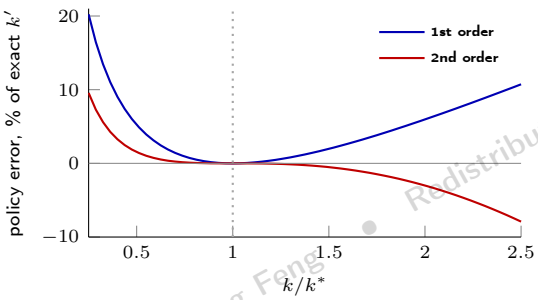
- Be accurate **far from the steady state**, or after a **large shock** (next frame).
- Cross a **kink**: an occasionally binding constraint has no Taylor series at the point where it starts to bind.
- Compute **global objects** — ergodic distributions, tails — that live where the expansion is poor.

Every limit is about **where** the approximation holds. Block B approximates on a whole **domain** instead, and pays in speed and dimension.

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Perturbation Is Local: The Picture

On Brock–Mirman in levels, the first- and second-order expansions are ordinary Taylor polynomials of a curved policy. Second order helps a lot nearby and little far away.



Brock–Mirman ($\delta = 1$) in *levels*: $k' = \alpha\beta k^{\alpha}$ against its Taylor polynomials at k^*.

In logs this model is exactly linear (5.A2), so no error could show there. In levels the policy is curved: zero error at the point, growing with the distance. Second order cuts the error 18-fold at $0.9k^*$ but only 2-fold at $2k^*$: **order buys curvature, not range.**

Example: error of k', %

k	1st order	2nd order
$0.5k^*$	5.2	1.5
$0.9k^*$	0.13	0.007
$2.0k^*$	6.0	3.0

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Ledger `pert_levels`; **errors formed from** `fig_exact`, `fig_first`, `fig_second`.

Everything so far expands around a steady-state point. In a heterogeneous-agent economy the state includes the wealth distribution, and the expansion point becomes the steady-state distribution.

Representative agent: expand around $(k^*, c^*, 0)$, one number per variable.

Heterogeneous agents (2.B1): agents differ in wealth and income, and the **cross-sectional distribution** Γ_t over wealth is part of the aggregate state. The expansion point is its steady state $\bar{\lambda}$ (2.A3): $\Gamma_t \approx \bar{\lambda} +$ (first-order deviation).

	Rep. agent → Het. agent
State	k_t (1 number) → Γ_t (N bin masses)
Steady state	$k^* \to \bar{\lambda}$ (N numbers)
Expansion	$\hat{k}_t \to \Gamma_t - \bar{\lambda}$

Idea

The logic does not change: Taylor-expand around a reference point that is no longer a scalar. If $F(\Gamma_t, z_t; \chi) = 0$ is an identity, differentiate it and match coefficients.

Notation

The literature often writes μ; this course writes λ for the invariant distribution, Γ for the distribution as a state, Q for the operator that moves it.

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3 · Perturbing Around $\bar{\lambda}$: Four Steps

Compute the stationary distribution, chop it into bins, linearize the law of motion around it, and apply Blanchard–Kahn to a state vector with hundreds of entries.

1: solve households at steady-state prices; iterate $\bar{\lambda} \leftarrow Q\bar{\lambda}$ to its fixed point ▷ 2.A3
2: discretize: a histogram with N bins, so the state is $(\Gamma^1, \ldots, \Gamma^N, z)$
3: linearize $\Gamma_{t+1} = Q_t\Gamma_t$ and the household policies around $\bar{\lambda}$ and steady-state prices
4: stack as $A\,\mathbb{E}_t s_{t+1} = Bs_t$; roots; check Blanchard–Kahn ▷ Part 1
5: **return** first-order impulse responses of a heterogeneous-agent model

The state vector now has hundreds of bins, but the problem is still *linear*, which modern linear algebra handles.

Idea

Same principle as Part 1: the distribution has more components than a capital stock, so A and B are large.

Example

Reiter (2009), *JEDC* 33, is this algorithm. **Auclert, Bardóczy, Rognlie & Straub (2021)**, *Econometrica* 89(5), organize the same idea around impulse responses. Session 6 builds the Aiyagari economy it applies to.

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Takeaways

- **Part 1.** Sort the variables into predetermined $x_t = (\hat{k}_t, z_t)$ and jump $y_t = \hat{c}_t$; the standard form $(x_{t+1}, \mathbb{E}_t y_{t+1}) = M(x_t, y_t)$. **Blanchard–Kahn:** as many roots of M above 1 as jump variables ($m_2 = n_2$, here 1); more gives no bounded path, fewer gives indeterminacy. Five steps: linearize; decompose $M = V\Omega L$; the canonical variables decouple; ruling out explosion gives the policy $C = -L_{22}^{-1} L_{21} = (0.590, 0.323)$; substituting back gives $K = M_{11} + M_{12} C$, whose roots are the stable ones $(0.962,\ 0.950)$. Undetermined coefficients keeps the same root; the phase diagram shows why only the saddle path converges.
- **Part 2.** Second order is 12 linear equations in 12 unknowns; the new term $\frac{1}{2} c_{\chi\chi}$ is a constant shift — precautionary saving, which first order reports as zero. Still local: second order cuts the error 18-fold at $0.9k^*$ and 2-fold at $2k^*$. **Order buys curvature, not range.**
- **Part 3.** The expansion point can be a **distribution**: stationary $\bar{\lambda}$, histogram bins, linearize, Blanchard–Kahn. Reiter (2009); Auclert et al. (2021).

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Bridge: Now Put a Number on It

Block A has claimed a great deal: fast, accurate near the steady state, blind to risk at first order, valid over a range it does not announce. Every one of those is a measurable statement about the quarterly model we already solved on a grid in 4.B4.

Block B measures them: 5.B4 grades this first-order solution against a global one, near and far from k^*.

Next (5.B1)

Projection: choose a domain and a family of functions, and make the Euler residual zero on it, not just at k^*.

And the question Block B answers: what do you do when "far from the steady state" is where the economics is?

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